

Three-dimensional Internet developments

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This paper describes the application of Internet-based three-dimensional (3-D) user interfaces, comparing them with more traditional two-dimensional (2-D) interfaces. The modelling of 3-D worlds in the Virtual Reality Modelling Language (VRML) is described, together with an overview of VRML and its likely future direction. Multi-user shared environments, which enable Internet users to meet and communicate with each other in cyberspace via avatars (3-D representations of users), are then introduced. A shared information garden is described which allows people to meet and share information in a dynamic, information-rich, virtual environment.

1. Introduction

We experience our physical world in three dimensions, through what we see, feel and hear; we are instinctively attuned to the 3-D world. Three-dimensional interactive multimedia information sources thus offer the possibility of an intuitive and natural way to explore complex ideas and concepts, and promote a quicker path to understanding. The ability to interact with on-line material through 3-D sight and sound can simulate the way we move about in the real world, giving us a sense of depth and perspective. In addition, it enhances the Internet navigation experience with a greater visual richness than the 2-D alternatives.

Virtual Reality Modelling Language (VRML) is a language for creating and viewing high-quality 3-D content on the Internet. There has been a considerable increase in the effort devoted to the development of virtual environments for human/human interaction and communication following the recent standardisation of VRML 2.0. In this paper, we discuss the motivation for using 3-D interfaces and the advantages which accrue from this approach. We then look in some detail at the VRML language, its history, current limitations and likely future directions. We discuss the use of World Wide Web (WWW)-based VRML in multi-user worlds and finally describe in some detail a novel WWW-based 3-D shared-space environment, the shared information garden.

2. Why use 3-D?

2.1 3-D and the information age

Three basic hypotheses regarding information storage and retrieval provide the motivation for the work described here:

- in the information age the production of information is easy — it is the ability to find and gain access to relevant information that is the key to success,
- since all data held in electronic form is digital, an iconic representation can be used to characterise attributes and relationships,
- browsing and searching the WWW and the Internet can be a frustrating experience, particularly for inexperienced users [1] — users can feel lost, disorientated and overwhelmed by long lists resulting from a simple search; information representation within the user interface using both 2-D and 3-D can reduce the time it takes to access information, make sense of it and enhance creative thinking.

A well-designed user interface with good spatial representation of information can be effective in assisting the user in the following tasks:

- browsing and navigation,
- searching,
- comparing,
- grouping,
- analysis,
- creating new information.

2.2 Why 3-D user interfaces?

Three-dimensional user interfaces are now a common experience, predominantly in the entertainment arena, and specifically 3-D computer games and simulators. Over the

last few years there have been a number of significant technical advances; these include the proliferation of powerful desktop computers, the ability to generate complex 3-D graphics and the increased connectivity to information and communication networks (particularly the WWW). This has opened the way to develop more meaningful ways of representing, and therefore communicating, information.

This section seeks to establish ways in which 3-D spatial user interfaces can be evolved to help people overcome some of the inherent problems with 2-D graphical interfaces and with the amount of information people now have available — the well-known ‘information overload’ problem.

The first user interfaces (UIs) were two-dimensional and text based. They were later superseded by graphical user interfaces (GUIs), with the advent of the Apple and Microsoft desktop systems. These interfaces drew on a spatial metaphor where the screen was seen as a desktop. Such metaphors made using applications simpler, easier, and more intuitive. It made computing more accessible and reduced the amount of training necessary to perform a useful task. Two-dimensional GUIs can offer clear and simple layouts; this is very useful for applications that display text or 2-D graphics. In such cases, a 3-D GUI may be ‘over-kill’ and detrimental to the task in question.

However, 2-D GUIs also have their limitations. Once multiple applications and files are open simultaneously, 2-D interfaces soon become cluttered and hard to handle. The cause of these problems is inherent in the nature of the desktop or windows metaphor, where layers are used over each other like papers on a table.

The two greatest problems with this type of interface are the limitations of space the users have available to arrange their work and the inability to gain a quick and accurate overview of the information contained on their system or in the network, and to navigate it quickly. This is particularly true in the case of 2-D WWW browsers. It is extremely hard to gain a sense of where you are in relation to the information you are browsing or viewing as there is no higher level overview representation of your position in the information space. Adding to your bookmarks list is the only way of creating a sense of what you are connected to and have stored. This gives the user little understanding or feedback as to the underlying attributes of the information, where it resides or how they found it. This is not an ideal way of delivering information, leaving no sense of location or navigation.

Three important areas have been mentioned; the use of metaphors, information overload, and navigation. Two-dimensional GUIs give the user 2-D spatial capabilities, i.e. if users wish, they can place two related folders close to

each other on the desktop. Such spatial organisation is very useful to the user. Many of the metaphor’s that the 2-D GUI use are partially reliant on spatial capabilities. The extra dimension brought to GUIs by 3-D also brings extra spatial capabilities. This in turn brings potential advantages.

‘We perceive and act in our environment on the basis of concepts of objects, persons, and events constructed, in large part, out of features and relations which are spatial.’ [2].

Waterworth [3] highlights some of the advantages of 3-D UIs. Spatial UIs are beneficial due to the natural human capacities for dealing with objects in space, behaving in space and attaching meaning to spatial dimensions, and the intrinsic memorability of spatially arranged information. Spatial representation of information capitalises on innate human capabilities for dealing with space and for precise sensori-motor control. Some of these advantages are looked at in more detail below.

Automatic behaviour — sensori-motor skills, with practice, can become automatic, like driving a car. They become largely unconscious, freeing attention so the user can focus on the reason for interacting, not on the process of interacting itself. However, automatic behaviour makes it harder to recall consciously. This could become a problem when navigating 3-D spatial interfaces. Therefore, provision needs to be made for memory aids, signposts, landmarks and distinctive features within the interface which can help remind users and locate them in space. Memory is essentially asynchronous communication with oneself. It is therefore important that the design of the spatial environment takes account of these human characteristics by providing these references to orientate the user.

Perceptual understanding — acquiring sensori-motor skills involves knowing **how** to go about doing something, rather than only knowing **what** is needed for the task [4]. These skills, and interfaces designed to exploit them, tend to foster an exploratory approach to content once the basic style of interaction has been mastered. This therefore encourages information exploration. Users should not be thought of as learning where things are; rather, they learn how to behave, almost unconsciously when looking for things. The key issue here for the interface designer is attaching meaning to the spatial arrangement of information within a 3-D user interface.

Semantics of 3-D UIs — using three dimensions of space can enhance display, navigation and access of information. However, 3-D instead of 2-D does not guarantee good usability. Norman [5] suggests spatial organisation of information only works when:

- there is a natural spatial mapping between (some, probably non-spatial, aspect of) the information objects and the spatial location,
- desired items can be located in a minimum of attempts,
- the number of different items at any one location is small enough that they can be readily found,
- the amount of work and time required to try a location, scan its content, and then try another location is small.

It is the first of these points that is problematic but potentially powerful. It is essential to attach meaning in a sensible way so users can purposely explore these objects and interpret shapes and locations easily and naturally in terms of the concepts in which they are interested. This is an example of the use of metaphors.

Mnemonics (memory aids) — it is plausible that spatial interface models for information systems can themselves serve as mnemotechnic systems (artificial memory) for their users. Mnemonic systems generally work by associating to-be-remembered items with various locations within an imagined 3-D world, like remembering the placing of a book in a room in a library. This is a classic memory technique — capitalising on imagery and motor-memory in location recollection.

Erickson [6] also supports the potential of 3-D spatial user interfaces (and indeed hints at their use in shared space environments which we consider in more detail later in this paper):

‘3-D seems to provide the possibility for representing many dimensions of information and meta-information in a compact and natural way. We believe that in the long run, information spaces will also be used as social spaces, and that 3-D can serve as a natural framework for supporting social interaction.’

2.3 Information retrieval in 3-D

In the information retrieval (IR) field, interest has grown recently in trying to improve the effectiveness of information retrieval systems by improving the interaction between user and system and in particular by designing various types of graphical front ends [7, 8]. In this view of IR, the role of the system is not to select documents similar to a user-supplied query but to organise and display information about many documents in such a way as to assist users to select useful documents on their own. It is hoped the following advantages will accrue from such an approach:

- users who are unfamiliar with the scope of a document collection will be able to familiarise themselves with it more easily,
- users who are unable to express their information needs in ‘traditional’ query formats will be able to locate documents more effectively,
- similarities between documents will be clearly presented which is not the case in a ranked list of documents matching a query.

Navigability in a purely symbolic domain has limits. The amount of ‘depth’ present in a subject before it exceeds human capacity for comprehension (and hence, navigation) is finite and relatively limited. Humans, however, have a sophisticated visual system and navigate in three dimensions; we are born to it and have a comprehensive ability to spatio-locate and spatio-organise. It thus seems reasonable to propose that the WWW should be extended, taking its navigation model from two dimensions into three.

2.4 Spatial design for information retrieval in 3-D

One of the most powerful features of 3-D graphics is that if the quality is sufficiently high then the interaction is sensually analogue. Instead of folders which are either open or closed a space allows information to be displayed in an infinitely variable granularity. With the increase in the power of agents able to synthesise and summarise information in a number of ways comes an increase in spatial metaphors allowing this information to be displayed in the same information space. This means we can move among information, changing the depth of our view without leaving our original context.

One of the main drawbacks when exploring information in 3-D is that it is not always straightforward to manoeuvre in a 3-D space using a mouse. This is often exacerbated by poor spatial design. This drawback can, however, be overcome by thoughtful design of the world. For instance in the information garden (see section 5), the information finding agent which generates the plants (3-D representations of information) also generates appropriate viewpoints from which to view those plants. Viewpoints are a facility supported by most VRML browsers which, when selected, automatically fly the user to a predetermined point in the space. One can imagine combining these viewpoints with a customer’s profile to generate a guided tour of the most appropriate information within a space. This also has an advantage — opening up the possibility of serendipitous information finding; for example, while being taken to see something an agent has identified as being important to you based on your profile, you may fly past some information which catches your eye. This is a specific example of a more general advantage of 3-D representations which users navigate — with careful design of the space, the journey around the space can itself have meaning.

3. Virtual Reality Modelling Language

3.1 Origins of VRML

The concept of a 3-D interface to the WWW was proposed by Parisi and Pesce at the first WWW conference in May 1994, with the concept of a 'Virtual Reality Markup Language' quickly gaining acceptance. By May 1995 the specification of the new language, now more correctly named 'Virtual Reality Modelling Language', had been defined [9]. This was based largely on a subset of the Silicon Graphics Open Inventor toolkit. The objective was for VRML to become the standard language for interactive simulation on the WWW. The first browser, WebSpace Navigator, appeared on the Silicon Graphics workstation at the same time the specification was published.

The first PC-based browsers appeared in the autumn of the same year, but the most significant step forward was in early 1996 when Netscape bought PaperSoft, one of the fledgling VRML start-ups, and included the PaperSoft VRML plugin with Netscape Navigator, naming it Live3-D. Suddenly everyone who had the Netscape browser also had 3-D capability — 3-D on the Internet took off.

Figure 1 shows a simple VRML 2.0 world, an office model, as viewed by the Cosmo Player browser.



Fig 1 VRML office model.

3.2 History and evolution of VRML

VRML is a textual language for describing 3-D worlds. Each separate object within the world is called a node. Nodes can be grouped together to form more complex nodes. VRML is a modelling language used to describe 3-D geometry and not *per se* a programming language (although the scripting capability of VRML 2.0 allows programming

languages such as Java to control behaviours and interaction in a VRML world).

VRML will develop in three stages, the first two having already been completed. The completed stages approximate to the first two versions of the language:

- defining appearance — how objects in cyberspace look (VRML 1.0),
- defining behaviour — how objects move and interact (VRML 2.0),
- defining distribution — how objects are shared (VRML 3).

The early days of VRML were driven by the VRML Architecture Group (VAG) which first met in August 1995. The VAG was set up by eight technical experts in the field, and was responsible for rapidly specifying and pushing through the first two versions of the standard. The VAG was dissolved with the formation of the VRML consortium.

The VRML consortium was founded in December 1996 as a non-profit organisation, whose mission is:

'fostering and evangelising VRML as the open standard for 3-D multimedia and shared virtual worlds on the Internet' [10].

3.3 VRML 1.0 — defining appearance

The specification for the first version of the language, VRML 1.0, was published in May 1995 [9]. This provides a means of creating, viewing, and (to a very limited extent) interacting with, static 3-D worlds.

Although spawned by the huge success of HTML, the VRML language is not an extension of HTML, since the latter is primarily for description of a 2-D WWW page, and the needs of a 3-D language are quite different from those of a 2-D language.

A practical decision was made early in the specification process that the only interactive behaviour in the first version would be hyperlinking, a direct analogy of the 2-D HTML links. This would allow hyperlinked objects, when activated, to link to any other WWW document (including another VRML world or HTML page).

Worlds in VRML 1.0 can be built from some 36 nodes defined in the language. The most significant of these nodes are as follows.

3.3.1 Shape nodes

Shape nodes (see Table 1) are the primary building blocks for worlds, and are used to describe the physical geometry of nodes.

Table 1 Shape nodes.

Node	Behaviour
Primitive shapes	Cone, cube, cylinder, sphere
Point sets	Set of points in 3-D, which can be joined by either lines or faces
Text	2-D text strings
Inline	Insert a VRML world, described by its URL, from elsewhere on the WWW

3.3.2 Property nodes

Property nodes (see Table 2) describe shape nodes, and how they relate to each other.

Table 2 Property nodes.

Node	Behaviour	
Appearance	FontStyle	Font descriptions for text
	Texture	2-D images to map onto surfaces of shapes
	Material	Description of shape material (colour, transparency, shininess, etc.)
	Level of detail	Allow different copies of objects at different resolutions
Grouping	Group nodes together, and transform them in space by rotation, scaling and translation	
Anchor	Add hyperlink to an object	
Cameras	Define position and type of camera used to view the world	
Lights	Light the world with directional, point or spot light	

3.4 VRML 2.0 — defining behaviour

VRML 1.0 was an excellent starting point for 3-D on the Internet. All the requirements needed for defining a static world were present. Users could navigate the static world, and hyperlink to other 3-D worlds (or 2-D documents), but no other interaction was available.

At the start of 1996, the VRML Architecture Group called for proposals for the second version of the language. This next version would be dynamic, allowing objects to interact with each other and the user.

Six proposals were put forward, the calibre of the companies involved in the proposals (such as IBM, Apple, Microsoft, Sun and Silicon Graphics) indicating the importance attached to these new 3-D developments in the computing industry. Discussion and voting was carried out within the forum of the thriving VRML mailing list [11]. Unsurprisingly, given that VRML 1.0 was driven by Silicon Graphics, their Moving Worlds proposal received over 70 percent of the votes cast in the poll in February 1996.

The Moving Worlds specification was updated by the VRML community, primarily by means of discussions on the VRML mailing list, and was published in final form as VRML 2.0 in August 1996 [12]. A growing number of VRML 2.0 books are becoming available; for a general overview of VRML 2.0 the definitive book is 'The VRML 2.0 Handbook' [13]; for tutorials and worked examples the 'VRML 2.0 Sourcebook' [14] is recommended.

Browser production for VRML 2.0 has not proceeded at the same speed as for VRML 1.0 because of the complexity of the specification (for interest, the VRML 1.0 specification runs to 25 pages, VRML 2.0 to over 200). The most widely supported platform for browsers is the PC running Windows NT/95, with early browsers being produced by Silicon Graphics and Sony. Netscape released a beta version of their VRML 2.0 Live3-D browser at the end of 1996. Although fully parsing the standard, no browsers yet fully implement the whole standard.

3.5 VRML 2.0 — extensions and enhancements to VRML 1.0

The VRML 2.0 standard allows worlds to become dynamic, defining how objects move and interact with each other and the user within the world. Although recognisably of the same heritage as VRML 1.0, the scope and capabilities of VRML 2.0 are much more extensive. The primary enhancements added by VRML 2.0 are given below.

- Enhanced static worlds — new nodes allow backdrops for the ground and the sky to be defined, and fog to dim distant objects. Nodes can be defined as billboards, so that they constantly face the user. Three-dimensional spatial sound has been added together with video (MPEG) textures.
- Interaction — sensors now allow the user to interact with objects by clicking them, dragging them or just being in close proximity to them. Time sensors allow objects to change over time, or actions to be triggered by a clock. These can be used to control animations.

'Collision detection' prevents the user from walking or flying through objects, and 'terrain following' allows the user to follow steps or ramps automatically.

- Animation — animations allow objects to follow predefined paths, or change colours. As animations can be compounded, a walking human body, for instance, can be created.
- Scripting — scripts allow the VRML developer to insert arbitrary program logic into a chain of events in the VRML world. So for instance clicking a light switch can turn on a light in the world.

Scripting is the power-house of VRML 2.0, providing the capabilities to make the world come alive. Scripting languages are not mandatory within the VRML 2.0 standard, but protocols are provided by the standard for Java and JavaScript.

- Prototypes — groups of nodes can be encapsulated into a new node type, and then reused within the world or by other worlds. Instances of the new type can be created with different field values. For example, a prototype could define an office desk. Instances of the prototype could then be a different coloured desk, or a desk scaled to a different size.

3.6 Authoring tools

Generating content for VRML can be done by hand in a text editor but it becomes time-consuming and laborious to construct all but the simplest of worlds. The only way to build more complex worlds is to use a GUI-based authoring tool. The VRML 1.0 language was quite simple in comparison to VRML 2.0, but suffered from a paucity of authoring tools. Authoring tools for VRML 2.0 are more prevalent and generally better but only a few allow a quality of control and manipulation essential for good VRML content.

Although visually appealing, worlds produced by even the best authoring environments can sometimes be inefficient. Because of this, experienced VRML world creators will typically use one of the authoring tools to build the world, and then 'tweak' the world by hand in a text editor to improve efficiency and add more complex animations or scripting.

3.7 Representation of 2-D text in 3-D

The representation of 2-D text in a 3-D environment is difficult from both the technical and design perspectives.

From the technical perspective it is difficult and slow to render text, because of the vast number of polygons which are required to adequately represent text characters. It is often more efficient to prepare a texture map representing the text off-line in a 2-D drawing package, and then map the texture into an object in the 3-D space. However, this is not generally feasible for dynamically generated text. From the design perspective, it can be difficult to position the text in a

meaningful way, especially when the text string, its location or its orientation are generated dynamically.

The use of 'billboards' in VRML 2.0 can ameliorate these problems to some extent by enabling the text to constantly face the user, whatever the user's orientation.

3.8 Navigation in 3-D

Navigation in 3-D is in one of a number of modes, selectable by the browser. The default navigation mode can also be set within the world:

- walk — used to explore the virtual world, the user generally being given a notion of gravity,
- fly — similar to 'walk', but with no notion of gravity,
- examine — used to view individual objects, including the capability to spin the object and move it closer or further away.

3.9 External Authoring Interface (EAI)

For communication between a VRML world and the external environment a standardised interface is required. The VRML 2.0 standard at present has no such interface. A proposal from Silicon Graphics [15] has gained widespread acceptance, and is expected to form the basis for the standardised interface. This interface allows external programs (currently only Java applets) to gain access to the nodes of a VRML world. This allows an external program to add, delete or change nodes in a world. Figure 2 shows an architectural view of how the EAI integrates with a VRML plug-in.

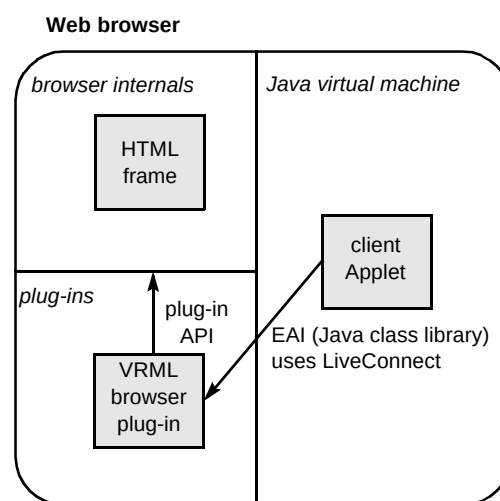


Fig 2 Architecture of External Authoring Interface.

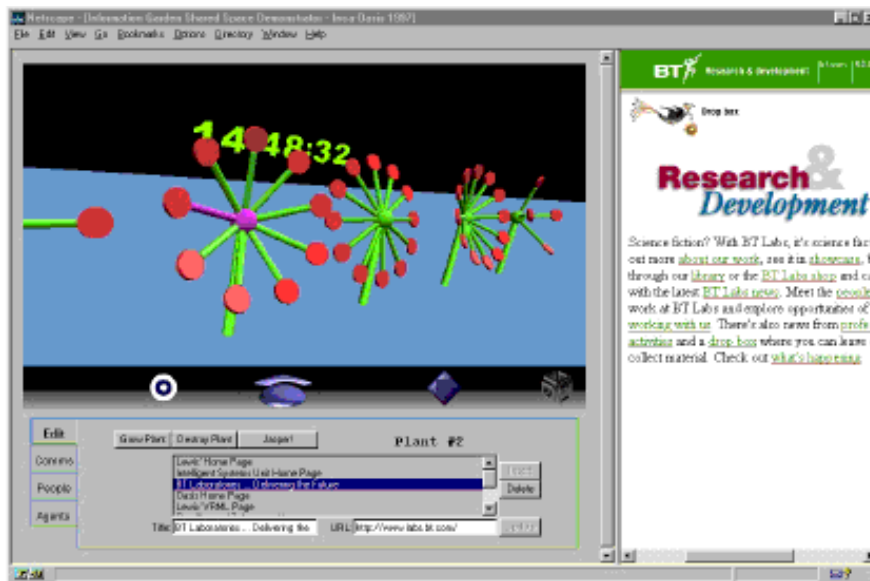


Fig 3 Local bookmark garden.

3.9.1 Local bookmark garden using the EAI

An example of the use of the EAI has been the construction of 'bookmark garden', a 3-D space representing a user's bookmarks, as shown in Fig 3. The window contains three sections; a VRML frame, a Java applet and an HTML frame. Each grouping of bookmarks is represented by a plant, with each bookmark a stem within the plant; the flower is an anchor (hyperlink) to the referenced document, which is presented in the 2-D frame. Stems can be selected either within the 3-D world or within a 2-D description in the Java applet. The applet allows the user to create or destroy plants dynamically, or to prune unwanted stems. Changes to the world are made dynamically under the control of the applet.

3.9.2 Multi-user shared world using the EAI

A more complex example of the use of the EAI would be to create a multi-user shared world. An external server would manage the location of each avatar (virtual representation in 3-D space of the user) in the world, and manage communication between avatars. On each browser, avatars representing other users are inserted into and moved within the world using the EAI. Users can bring their own 3-D objects to the world and make them shareable by other users. Persistence in the server would enable new users to enter the shared world as currently seen by other users (which may of course be considerably different from the original world). Figure 4 shows a possible architecture for such a system. Such a demonstration system has been built at BT Laboratories and section 5 discusses this application in more detail.

3.10 World optimisation

To be acceptable for a user, a world must both load and navigate quickly. User impression of the speed and usability of a world is dependent on two major factors.

- **Size of world** — download time over the WWW significantly affects user perception of the world. While hand-built simple worlds can be very small, the first generation of authoring tools were notoriously inefficient at producing VRML. Intelligent design of worlds can radically reduce their size without compromising the quality, for instance, reusing previously defined objects such as prototypes. The information garden (see section 5) was reduced in size by a factor of ten by use of prototypes to define parts of the plant stems.

VRML is at present defined as a textual language, and as such can be easily understood by authors, and the source of the world viewed with a browser. Worlds can optionally be compressed by gzip (a no-loss compression which will typically compress the world by a factor of between five and ten). Most browsers will decompress on the fly, with the time saved on transmission of a compressed world far outweighing the time required for decompression.

Plans are under way for a binary representation of VRML [16], based on IBM's geometric compression format. This will not only reduce download time, but will also reduce the time required by the browser to parse the VRML.

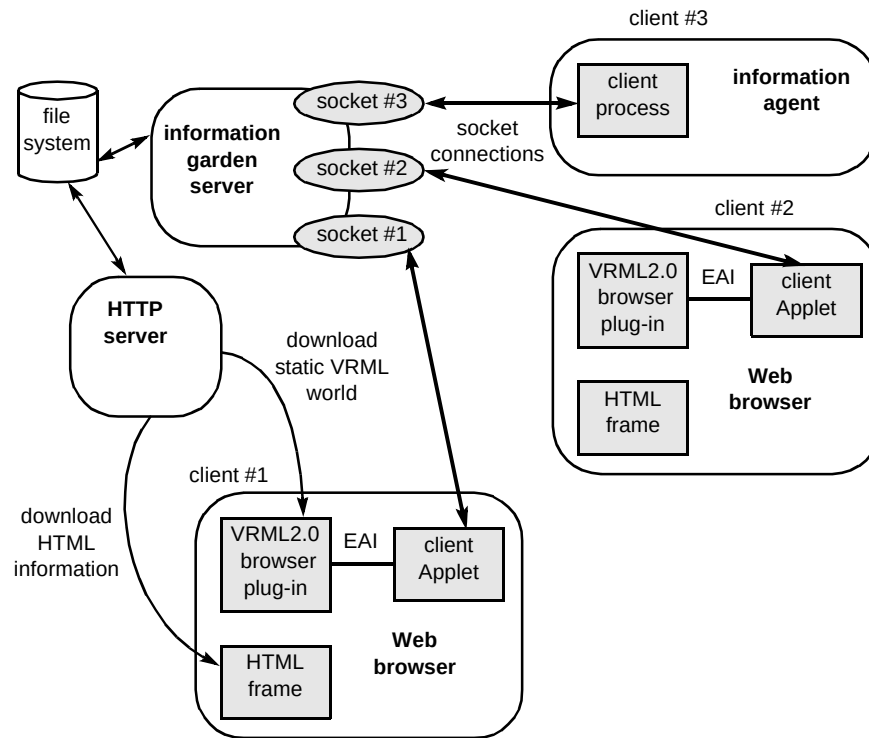


Fig 4 Multi-user shared world using EAI.

Waldrop [17] produced a simple utility, the Datafat Munger, to shrink a world by removing redundant information (such as rounding floating point numbers and eliminating white space). BT's 'vermel wormel' compressor is based on this but with improved performance.

- **Improving rendering** — to make a world feel interactive, a minimum frame rate of 10 frames (images) per second is required, although 30 frames per second is ideal. While the user can improve the frame rate by reducing the window (display) size, much can be done by the author by the efficient design of worlds, such as reducing the number of polygons used to model objects (polygon reduction), using 'levels of detail', and intelligent use of textures.

The use of standardised rendering engines (such as Microsoft's Direct3D), together with the imminent launch of 3-D PC cards to support these engines will cause a significant improvement to the rendering speed of worlds.

The above two factors are in turn dependent upon five significant technical advances which are currently being made. Each of these will improve the perceived usability of VRML worlds:

- improvement in PC performance and memory (doubling every 18 months),

- 3-D rendering engines and associated 3-D cards,
- browser performance improvement,
- authoring tool improvements (to create more efficient worlds),
- binary representation of VRML.

3.11 Live data

Although VRML is primarily a 3-D world definition language, with worlds served by a WWW server via HTTP, there is no need for the world to exist already within a file on the WWW server. Scripts on a WWW server can generate VRML 'on the fly', in much the same way that HTML pages can be created dynamically. Scripts such as shell scripts or PERL scripts can create the world when requested by the browser. As a simple example, one of the avatars used in the information garden is created dynamically from the BT share price — the rise or fall of the share price during the day is reflected in the size of the avatar's smile or frown.

Because worlds can be built from a number of components using the inline node, worlds can be built using a combination of pre-existing and dynamic sub-worlds.

Such dynamic data can add considerable value to a world. Just as 2-D pages are increasingly being dynamically created (such as with data from databases), it is expected

that the dynamic creation of 3-D worlds within VRML will find considerable use for information navigation.

For worlds to be fully dynamic, dynamic creation is not sufficient — the world itself needs to be updated in real time. VRML 2.0 allows nodes to be dynamically added to or deleted from a world, without having to re-render other nodes, by use of the EAI (section 3.9). This allows parts of a world to change to represent some external conditions. For instance, a user viewing a 3-D view of a file store will see representations appearing or disappearing in the world, corresponding to files being created or deleted in the file store.

3.12 Streaming of information (data, audio, video)

The VRML 2.0 standard made worlds come alive by the inclusion of audio and video. At the present time these are stored files, transferred across the WWW when the world is loaded, and cannot be rendered in the world until the whole file has been received from the WWW server.

With the use of the EAI, it will soon become possible to stream audio and video into a world. So, for example, a world could have a radio station playing live as ambient audio, or a TV could be showing live TV (and could change channels by pressing the appropriate channel button on the model of the TV).

The streaming of audio and video over the Internet is covered in Rudkin et al [18].

3.13 International standards

As with many standards which rapidly evolved from industry initiatives, the VRML community realised that the VRML standard needed a more sound international definition. This was seen as crucial for its growth and international acceptance. At its July 1996 meeting in Kyoto, Japan, ISO's JTC1/SC24 committee agreed to publish VRML 2.0 as Committee Draft (CD) 14772. The CD ballot closed in November 1996, and the Draft International Standard (DIS) was made available in February 1997; progression to full International Standard (IS) is anticipated for August 1997. The ISO standard will be called VRML97 (with 'modeling' now spelt 'modelling').

3.14 VRML 3 — Defining distribution

The VRML 2.0 standard does not define methods for distributing the world on multiple clients and thus making the world multi-user. Such multi-user capabilities can however be achieved using non-standard methods with the EAI, as discussed in section 3.9 above, or by adding proprietary nodes to VRML 2.0.

No significant work has as yet been undertaken to define VRML 3. If and when it is created, the following are likely candidates for inclusion:

- binary file format,
- standardisation of external API,
- language definition for script nodes — there are currently no mandatory languages,
- multi-user capability,
- seamless worlds — where worlds can be joined automatically,
- heads-up displays — to allow nodes to be locked to the camera,
- full inter-object collision detection.

4. Shared environments (multi-user worlds)

While the first stage of VRML introduced static worlds and the second moving worlds, it is the third stage which may prove to have the most impact. This third stage introduces distributed worlds, enabling users to meet, interact and communicate with each other in a shared world across the Internet.

4.1 Avatars

In a typical shared 3-D environment, users are able to choose their avatar (their virtual representation in the shared environment) from either a standard set of avatars provided by the product, or by defining their own avatar, usually as a VRML model. Although typically defined by a VRML world, there is no need for such worlds to be static. Avatar gestures, such as waving or smiling, can sometimes be selected in real time by the user. Additionally avatars can be generated dynamically; so in the information garden (section 5) there are avatars whose appearance depends on the current rise/fall of the BT share price, and avatars dependent upon IP address (for instance, an avatar can appear in 'casuals' if its owner has entered the environment from home rather than the office PC).

As users navigate a world, their avatar can be seen by other users of the same world. As they move, so their avatar is seen to move by other users. Example avatars [19] are shown in Fig 5.

4.2 Commercial shared environments

The earliest shared environments, such as World's Chat [20], were based on proprietary browsers, worlds and multi-user servers. The CyberGate browser from Black Sun [21]

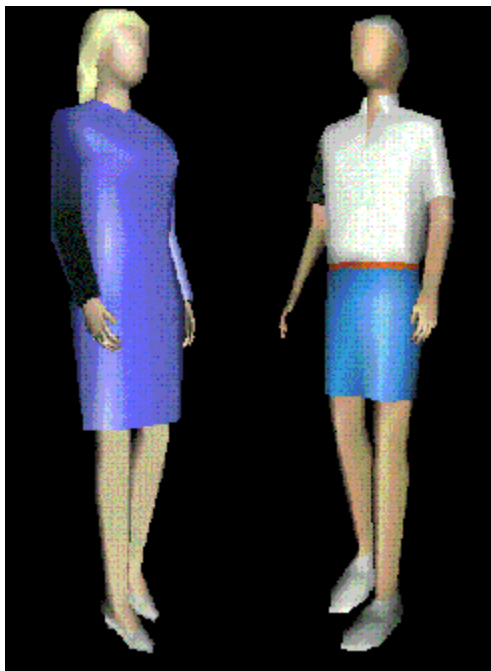


Fig 5 Example avatars [19].

launched in early 1996 used VRML 1.0 with extensions to define the worlds. Sony's Community Place [22] was launched soon after the publication of the VRML 2.0 standard, and uses VRML 2.0 for its worlds with a Java applet driving the multi-user capabilities of the client; avatar tracking and communication is handled by a server.

At present none of the shared environments interwork. It is crucial to the success of shared environments that international standards are defined and implemented for their interworking. At the current time a number of initiatives are under way, the most notable being Living Worlds [23].

4.3 The personalisation/sharing dichotomy

The sharing within multi-user worlds has an interesting and unfortunate ramification for the use of information filtering. If information within the garden is filtered for a particular user based on a profile of their interests, then shared objects may not be identical.

As an example, in the physical world one person may arrange the books in their bookcase by author's surname. A second person may order identical books by subject matter. In a virtual environment it is perfectly possible for both configurations to apply to the same bookshelf — books are simply arranged for each user according to their profile. Such an arrangement shows the power of virtual environments, since this is of course not possible in the physical world. However, making the virtual world shared produces inconsistency problems. To one user the book may appear on the top shelf, to the second user it may be on a lower shelf.

5. An example application — a shared information garden

The objective of the shared information garden project is to develop unified 2-D/3-D shared, dynamic and intelligent information interfaces and representations. The main focus of the work is the user exploration and sharing of WWW-based information resources through the user interface. The information garden should thus:

- generate an easy-to-use, engaging, unified information interface,
- support collaborative, communal interaction and communication,
- build live, self-updating information places,
- illustrate shared access to information places,
- support persistence in the multi-user server,
- provide communication (audio, video, whiteboards, etc) for shared-space user interfaces.

The garden metaphor was chosen for a number of reasons, in particular the fact that the concept of a garden is highly cross-cultural and also because it was felt that this metaphor reflected the organic, changing nature of information which it was hoped to reflect in the space.

5.1 Outline

The project has built on previous work and extended the spatial metaphor of previous information landscapes and shared spaces as a framework for the user interface. Previous work includes BT's Portal [24] and The Mirror [25].

The interface concept is defined, firstly, by the capability to navigate a self-generating hierarchical information structure from which the user can browse and view documents on WWW servers. This provides a constantly updating and persistent information window on the desktop, utilising a unified 2-D/3-D interface to its best advantage.

Secondly, the interface design allows a seamless transition to linked shared-information environments or 'places' from any layer of the WWW visualisation. The visualisation uses the metaphor of 'information constellations' through which to navigate and explore. The entry link from the WWW visualisation connects the user to a shared user environment. Here the user, represented as an avatar, can navigate directly to specific unit, project, personal or communal shared 3-D 'places' containing representations of context/activity-driven and user-defined information. This environment is supported by the Jasper information agent [26] which retrieves information that is then represented in 3-D. Within the shared environment,

users can interact and access WWW information including status information on user connectivity and communication facilities.

Figure 6 shows a draft sketch of the shared information garden 3-D space. There are three main levels in the space. At the top level is a 'solar system', whereby the structure of the solar system reflects the structure of the information being represented. Each node or 'planet' represents a set of WWW pages related to a particular department within a company (but could of course be any other natural grouping of information), so that the most central object could represent the company itself with smaller organisational units of the company (divisions, departments, groups) being indicated by distance from the centre. The user can navigate around this space until they 'zoom in' on where they need to be. Clicking on a particular planet then brings up relevant information in a 2-D WWW browser.

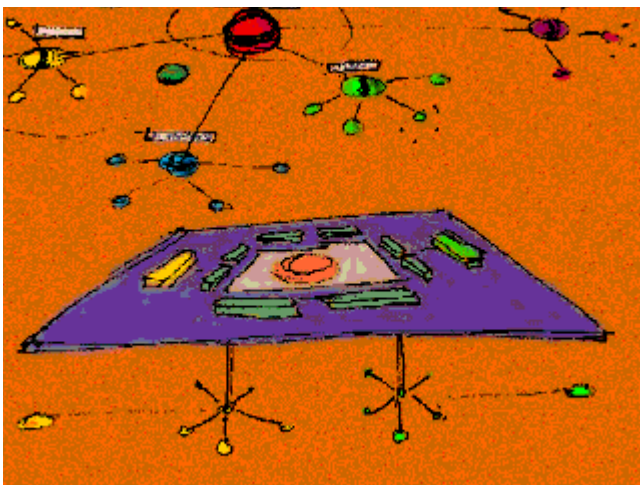


Fig 6 Shared information garden — top level view.

On the second level of the space is a shared information environment and this is discussed in more detail in the next section. The third level is reserved primarily for systems maintenance activities (which can themselves be carried out within the space).

5.2 Jasper shared information store

A major consequence of the introduction of WWW to the Internet has been an increase in the usability of the information within it. This is directly attributable to the navigability of the information — in other words, the Internet is useful (and will be used) to the degree it is capable of conforming to requests made of it. The WWW has greatly improved the accessibility of the Internet. Furthermore, it has added a universal structure to the information within it; through WWW, the Internet can be treated as a single, unified data source, and all of the information can be treated as a single, albeit highly complex, document.

Given the immense amount of information available on the WWW, when information of interest is found, it is preferable to avoid copying entire documents from the original location to a local server. The usual bookmark facilities provided by browsers are however limited, typically allowing the local storage only of the title of a WWW page and its uniform resource locator (URL). The Jasper information agent [26] is an attempt to address this problem. Users can 'store' pages of interest in Jasper. Jasper extracts the important details from these documents, including the summary and keywords, as well as the document title, its URL, and the date and time of access. This information is maintained locally and has several purposes. The summary gives a user an idea of the content of the document without the need to retrieve the remote pages. A quick link to the full remote information is also provided. Jasper locally indexes the pages that have been stored. Over time, a rich set of pages is built up by Jasper from different users' entries.

When a user stores a page in the Jasper system, that user's agent automatically informs other users with matching profiles, by e-mail. A reference is also added to every user's 'What's New?' page along with a predicted interest rating. Furthermore, if the document does not match the profile of the user who stored it, then Jasper prompts that user to update their profile accordingly and suggests suitable additional keywords. In this way Jasper learns more about the user's interests as the system is used.

5.3 Information representation within the garden

Within the shared environments the ability to generate WWW-based information representations is key. Agents like Jasper, described above, are used to provide shared-information resources and the ability to generate group- or personal-based searches and representations, providing another layer of information support inside the shared 3-D spatial interface.

It is proposed to exploit the pages stored in the Jasper system by making part of the 'information garden' a 3-D representation of the Jasper store, through and over which users can fly and wherein information is arranged in a logical, intuitive and accessible way. This arrangement of information is achieved by using the metainformation in the Jasper store and results from an automatic clustering process which groups together URLs (WWW pages) containing related information. The resulting cluster structures of the Jasper store are thus used to create a novel 3-D front end on to the Jasper system, using VRML. This work is described in more detail in Davies et al [8].

Figure 7 is a screenshot from a working prototype shared information garden (the second level of our shared information space described above and shown in Fig 6). The central circle shown in Fig 7 is in fact a shared area where

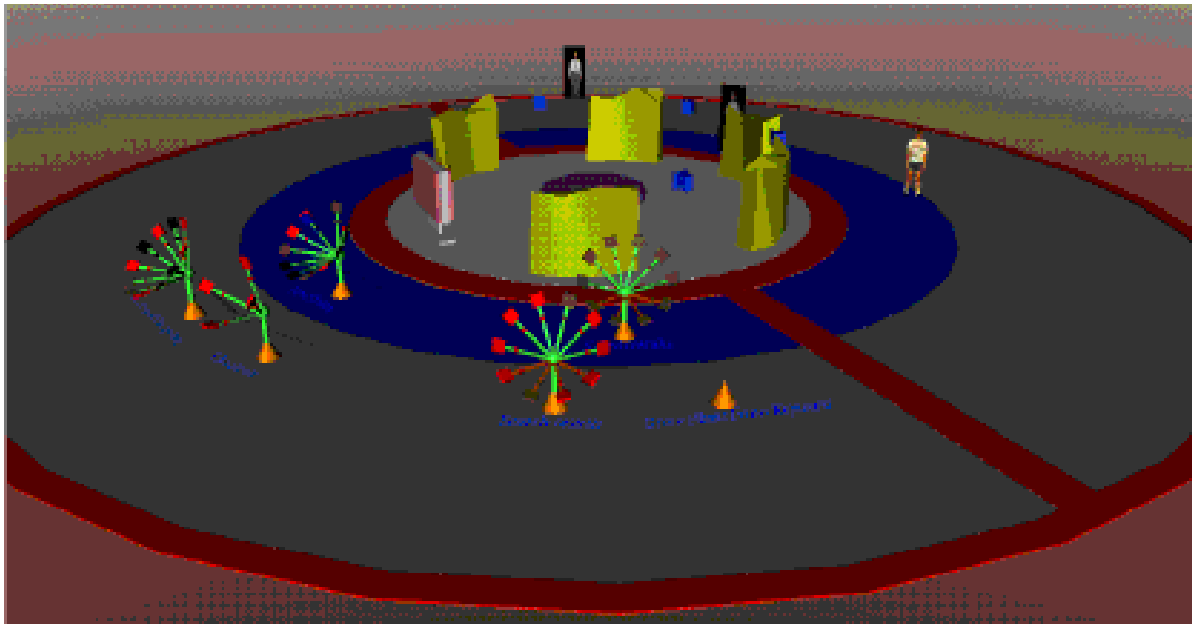


Fig 7 Shared information garden.

avatars can meet and communicate via a number of media including text, audio and video. Shared electronic whiteboards are also provided. The outer sectors are different areas of the information garden. Each sector represents a different information source. As the user navigates towards a particular sector, more detail becomes visible, as shown in Fig 8. Current sources of information supported are users' bookmarks, the Jasper information agent store (described in section 5.2 above) and a news feed sector, where plants grow and die as news arrives and then ages. Figure 9 shows a virtual shared newsroom currently being implemented. It is intended that this and other rooms will become special interest areas which users can enter from the central shared space.

In Fig 8, clustered groups of documents can be seen (data mining, distributed computing, etc). Each document is represented by a stalk with a coloured square at its end. The colour coding of the squares on the stalks represents the status of the particular document: red circles indicate that a document has been updated since it was stored in Jasper, black that the link is 'dead' (that is, the URL associated with this document is no longer valid), while a blue circle indicates that the document in the Jasper store is up to date. The smaller circles half-way up the document stalk represent a locally held summary of the information, as generated by Jasper. When a document is selected, the associated URL is loaded into a separate 2-D browser window. The selected stem then starts to wave, analogous to

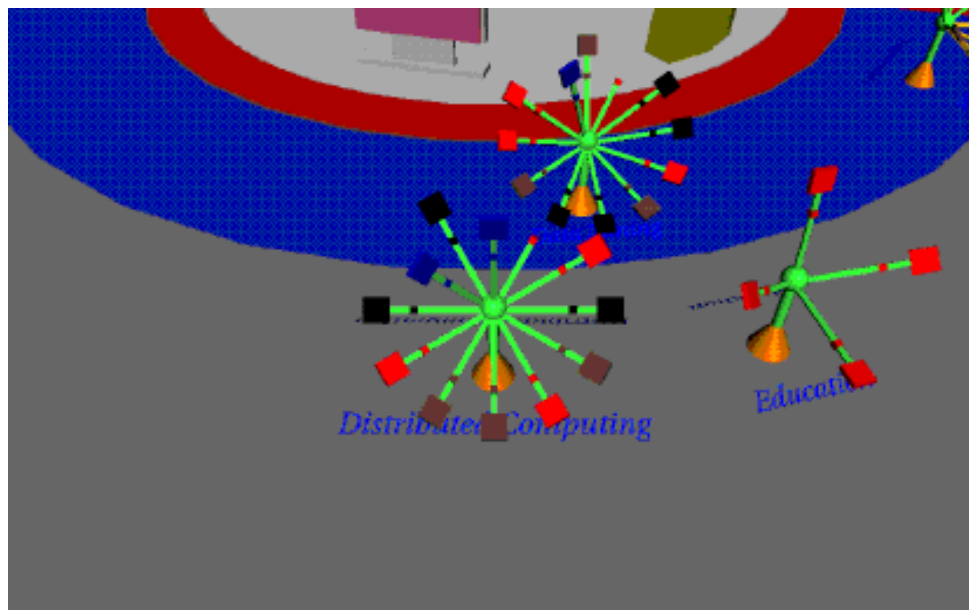


Fig 8 Information garden document cluster.



Fig 9 A virtual shared newsroom.

touching stems on a real plant as one walks through a real garden. The wave period decreases over time. This visual representation allows, for instance, a 'busy' or 'hot' plant to be immediately visible to other users by virtue of the number of moving stems.

At some point, it makes sense to drop down from a 3-D to a 2-D view of the information and, by clicking on the large or small squares, the user is shown a WWW page of the original information or the summary respectively.

5.4 Information sharing within the garden

By making the information garden multi-user, users can not only access the rich variety of information resources within the garden, but collaborate with other users across the Internet. The avatar representations of users are not just visual representations, but contain communications and profiling information about them. Thus when avatars meet in cyberspace, they can communicate via a rich variety of media. The communications channels are set up automatically within the garden. These include:

- text,
- speech (both Internet telephone and PSTN telephone),
- audio/video,
- diary scheduling,
- white board,
- application sharing.

Users can additionally bring their own 3-D objects, such as those representing further information sources, to the

garden. As an example, users can bring a copy of their personal Internet bookmark file to the garden, and allow other participants to view it and thus access the information referenced from within it. Taking the gardening metaphor further, users can take 'cuttings' from information plants and grow their own copy within their personal environment. When a cutting is taken, the document represented by the stalk in question is analysed and a query in the form of a set of key phrases is extracted from it. So, for example, if the document was about network-centric computing the query extracted might be:

{network-centric computing, java, network computers, mobile code, thin clients}.

This query is then passed to a search engine which searches WWW for documents which match the query. After some post-search filtering, the ten most relevant documents are then represented as stalks on a plant which grows in the 3-D space.

Furthermore, since each user within the shared environment has a profile maintained by their avatar, software agents can proactively bring together users who have similar profiles and enable information sharing between them. Alternatively, users endeavouring to research information on a particular topic not only have access to the wealth of links stored within Jasper and any other information sources represented in the garden, but can be automatically put in touch with (perhaps unknown) colleagues who have expertise in this field and meet them in the shared environment.

5.5 *Evaluation of the information garden*

The shared space is currently being trialled at BT Laboratories and early feedback from users is encouraging. Features to which they have reacted particularly positively include:

- the 'organic' nature of the garden metaphor, which encourages the idea of information which will grow (and be pruned),
- the ability to visually represent a reasonably large collection of documents at different levels of detail,
- the metainformation about the information collection provided by clustering and colour coding of the documents, particularly when viewed at relatively long distance — examples of the information provided include the amount of information available on a given topic and clusters where many links have 'died' or need updating,
- the tendency to get 'lost in hyperspace' is reduced, since the user can stay in the same 3-D space and yet have access to a range of information and meta-information at multiple levels of detail.

Development of the garden is continuing in the light of user feedback.

6. Conclusions

The use of 3-D on the Internet has opened the way to the development of more meaningful ways of representing, and therefore communicating, information. Three-dimensional spatial user interfaces can be created to help people overcome some of the inherent information overload problems with 2-D graphical interfaces. It is anticipated that the use of 3-D for information representation, information sharing and virtual social interaction will increase rapidly.

VRML provides a cross-platform international standard for the representation of such 3-D worlds, and is now widely supported by browsers.

VRML content is already proliferating on the World Wide Web in virtual chat worlds, virtual neighbourhoods and on-line shopping malls. Other application areas range from remote microsurgery to 3-D games, from virtual offices to 3-D on-line educational packages and more. For the future, a vast community of 3-D environments is envisaged, with rich social interaction and immersive sensory experiences.

A number of developments, both directly and indirectly related to 3-D, are expected to significantly increase the use of 3-D on the WWW.

- General technology trends — the hardware price/performance ratio will continue to decrease and bandwidth will become significantly wider and cheaper. Faster rendering (by both hardware and software) will provide more realistic worlds.
- Further standardisation within 3-D — there will be further standardisation of scripting, avatars and shared spaces. The visual differences between a world as rendered by different browsers will diminish, and there will be better cross-platform interworking.
- World building and viewing — authoring suites will become more powerful, and browsers will become more efficient and implement more of the standard. Heads-up and stereo display will provide richer immersive environments.
- Usability of 3-D worlds — worlds will become richer and more realistic. They will become more organic, not only being built dynamically, but maintained and updated dynamically. Greater personalisation of the worlds will be provided by the use of profiles. Within the worlds, greater interactivity with objects will be provided, together with a tighter integration with relevant 2-D information.

Acknowledgements

The authors gratefully acknowledge the work of Lewis Collins of BT Laboratories on the EAI.

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He is currently involved in the dynamic representation of information in 3-D on the Internet, in particular with VRML and shared spaces.



Andrew McGrath leads a team of designers and animators at BT Laboratories working on shared spaces projects. His most recent work has been on The Mirror — a collaboration between BT, Sony and the BBC, exploring inhabited TV.

Since leaving college in 1991, he has focused on 3-D user interfaces, including management of the Portal — an early VRML site. He has a BA in 3-D Design from Glasgow School of Art and an MA in Product Design from Manchester Polytechnic.